MILITARY SPECIFICATION

SIMULATOR, TARGET: 10558244

1. SCOPE

1.1 This specification covers a target simulator, a part of Range Computer Test Panel, Subassembly MX-8049/TPM-22 a part of Radar Test Set AN/TPM-22.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Amy Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-c-46357	Computer, Gun Direction: M18; Manufacturing Process and Inspection of

STANDARDS

Military

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MIL-STD-105	Sampling Procedures and Tables for Inspection *by* Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

F10554567	Schematic, simulator, Target
F10558244	simulator, Target

PACKAGING DATA SHEET

10558244	Packaging of Simulator Target

(Copies of specification, standards, drawings, and packaging data sheets required by tile suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3.1 Fabrication. - The target simulator shall be manufactured in accordance with Drawing F10558244 and drawings pertaining thereto.

3.2 General Specification. - Where applicable, the target simulator shall meet the requirements of MIL-F 13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13326

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C--46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners

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d. Marking

e. Finishes

3.3 Enviromental service conditions.

3. 3.1 Vibration. - The target simulator shall be mounted on the vibration table in normal service position. The target simulator shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps\sec) . Vibration tests shall be conducted under ambient room conditions.

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3.3.2 Shock. - The target simulator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks *in* opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 + 1 milliseconds. The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. - The target simulator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 nonoperating. - The target simulator shall meet the requirements of 5.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. - The target simulator shall meet the requirements of *3.4* and 3.5 when stabilized thermally at any temperature between +32° F and +131°F.

3.3.4.2 Nonoperating (storage). - The target simulator shall not suffer permanent damage when subjected to the temperature extremes of -80° F and +185° E', and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating. - The target simulator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. - Each conductor shall be continuous electrically as specified on the schematic diagram F10554567

3.4.2 Resistance. - Resistance shall be 100 ohms ±10% between the following terminals.

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FROM	TO
Pi-l	P1-2
P1-5	P1-7
P1-18	P1-19
P1-20	P1-22

3.5 Performance. - When supplied with the inputs specified in Table I, the target simulator shall yield the outputs specified in Table I.

TABLE I. - INPUT AND OUTPUT CHARACTERISTICS

REQUIREMENT	CHARACTERISTICS	TERMINALS
Input power	5.0 ±0.3 Vdc regulated, ac ripple net greater	P1-30 (+) p1-24 (Ret)

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TABLE 1. - INPUT AND OUTPUT CHARACTERISTICS (Cent'd)

	than 50 mV peak-to-peak 300 mA capacity	
	33 ±1 Vdc, ac ripple not greater than O.lV peak-to-peak, 10 mA capacity	P1-5 (-) P1-25(Ret)
Input signals		
Signal #l	positive pulse from zero volts to not less than +4 Vdc, 0.5 ±0.1 usec wide, not greater than 40 nsec rise time, rep. rate 20 kHz ±200 Hz	P1-19 and P1-18 (Ret)
Signal #2	positive pulse from O V to not less than +3 Vdc, 0.35 ±0.05 usec wide, triggered by Signal #l	P1-20 P1-22 (Ret)

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TABLE 1. - INPUT AND OUTPUT CHARACTERISTICS (Cent'd)

Output signals		
Simulated target	0.2 ±0.1 usec positive pulse from O V to net less than +3 Vdc, delayed 2 .to 30 usec from Signal #l. Delay controlled by resis-tance between J1-42 and J1-47 and proper switching between J1-33 through J1-38	P1-16 and P1-14 (Ret)
Sum doppler	2.5 kHz ±1% square wave from O V to not less than +3 Vdc	P1-9 and P1-10 (Ret)
Difference doppler	2.5 kHz ±1% square wave, from O V to not less than +3 Vdc, in phase	P1-12 and P1-8 (Ret)

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TABLE I. - INPUT AND OUTPUT CHARACTERISTICS (Cent'd)

	with sum doppler when simulated target appears before Signal #2

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility <Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one target simulator manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the -contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual charscteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and

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Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE II - CLASSIFICATION OF DEFECTS

100% INSPECTION

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2	4.6.2.2
103.	Performance	3.5	4.6.3

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling - one target simulator out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE III - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specification	3.2	MIL-F-13926 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3

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TABLE III. - SPECIAL SAMPLING (Cent'd)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306. Altitude - nonoperating	3.3.3.2	4.6.1.3
307. Temperature	3.3.4	4.6.1.4
308. Humidity	3.3.5	4.6.1.5

4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-1-45607.

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4. 5.1 Government- furnished inspection equipment - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of NIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for -which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when

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nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

4.5.2.2 Test equipment. - In conjunction with 4.5.2.1, the following Lest equipment shall be used in the performance of the applicable tests as specified in 4.6:

Item Number	Description
1. Ohmmeter	To measure 100 ohms ±10%
2. Power supply	5 Vdc ±0.3 Vdc regulated, ac ripple less than 50 mV p-p, capacity 300 mA
3. Power supply	33 Vdc ±1 Vdc regulated, ac ripple less than 0.1 V p-p, 10 mA capacity

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Item Number	Description
4. Signal generator	positive pulse, O V to no less than +4 Vdc, 0.5 ±0.1 usec wide, not greater than 40 nsec rise time into 50 ohms , 20 kHz repetition rate
5. Signal generator	positive pulse, O V to no less than +3 Vdc, 0.35 ± 0.05 usec wide; externally triggered at 20 kHz
6. Oscilloscope	dual channel; rise time of scope plus probe not greater than 10 nsec., sweep period not less than 50 n sec.
7. Potentiometer	10 k ohms, 10 turn

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Item Number		Description
8.	Switch	rotary, three position
9.	Connector, receptacle	per Frankford Arsenal F10554938 or equal

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-81O except as specified herein. Testing equipment used in environmental tests shall be in accordance with “Test Facilities and Apparatus” requirements of MIL-STD-81O.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure criteria of 4.9.1.3 does not apply. After vibration, the target simulator shall be tested as given in the subparagraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

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4.6.1.2 Shock. - This test shall be performed as speci - fied in Procedure I of Method 516 of MIL-STD-81O except that shock pulses shall be 11.0 ms and 15 g's. After the shock irnpulses, the target simulator shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2,

4.6.1.3 Altitude. - non-operating and operating. This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-81O. The temperature of Step 2 shall be +185°F . The temperature of Step 4 shall be +130°F. In Step 5 the target simulator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating) . - This test shall be performed as specified in Procedure I, Method 502

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of MIL-STD-810. The temperature of Step 2, shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the target simulator shall be tested in accordance with 4.6.2 and 4.6.3

4.6.1.5 Humidity-nonoperating. - This test shall be performed in accordance With Procedure I, Methods 507 of MIL-STD-81O except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical.

4.6.2.1 Continuity. - Continuity shall be inspected by standard measuring equipment (WE) and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance. - Resistance shall be measured with an ohmmeter, item 1 for compliance with 3.4.2.

4.6.3 Performance

4.6.3.1 Simulated Target. - Connect test equipment item 1 through 4 and 6 through 9 as shown in figure 1. Adjust

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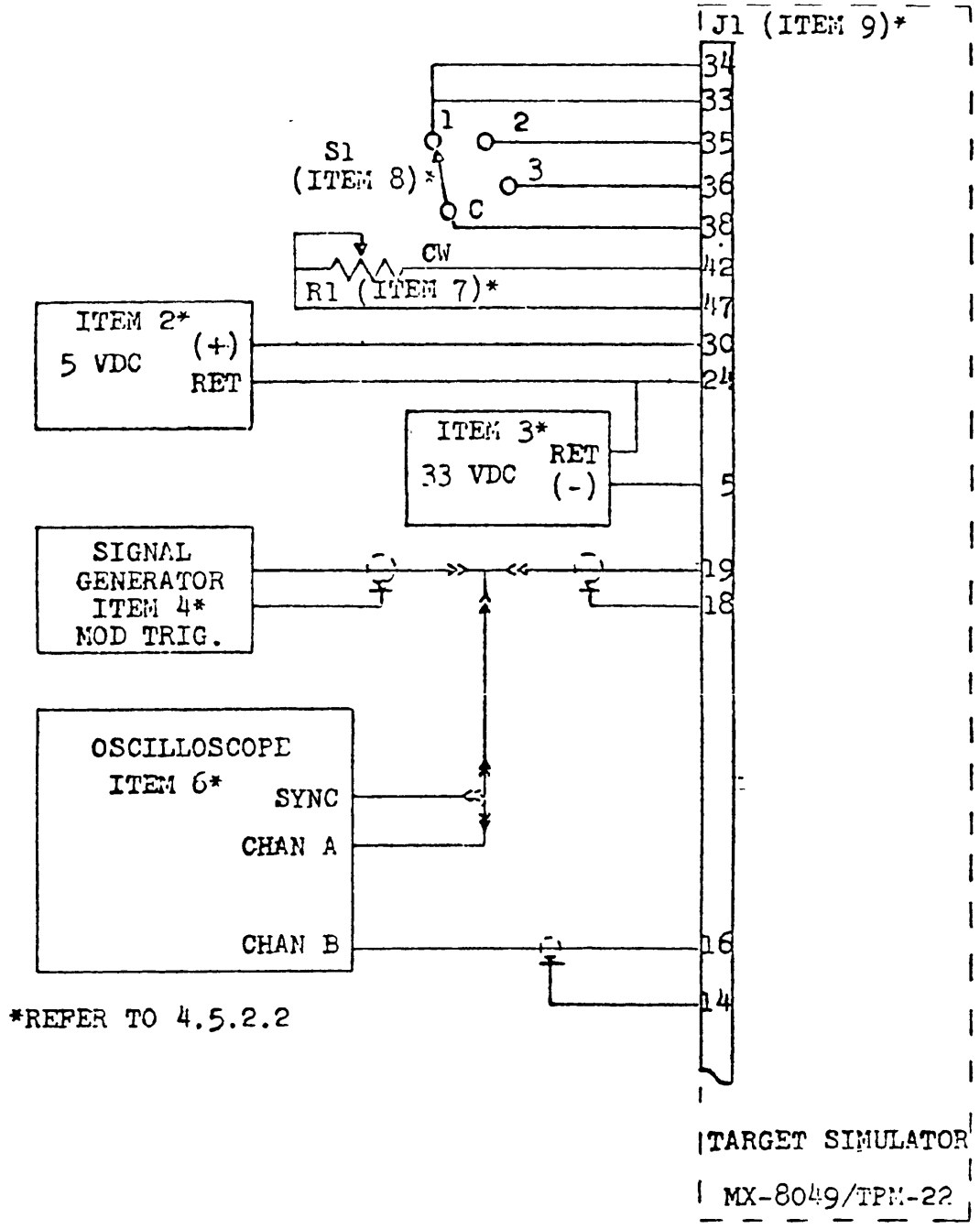


FIGURE 1. SIMULATED TARGET TEST SET UP

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signal generator item 4 to deliver Signal #1 as specified in Table I and perform the following steps.

(a) Set S1 to position 1.

(b) Adjust RI from fully counterclockwise (CCW) to clockwise (CW) . Verify that simulated target pulse appears on oscilloscope channel B and verify that the time between leading edge of signal on channel A and leading edge of signal on channel B varies from 2 to 10 usec with adjustment of R1.

(c) Set R1 to fully CCW.

(d) Set S1 to position 2.

(e) Adjust R1 from fully CCW to CW. Verify that simulated target pulse varies from 10 ±1.0 usec to 20 ±1.0 usec after signal on channel A.

(f) Set RI to fully CCW.

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(g) Set S1 to position 3.

(h) Adjust R1 from fully CCW to CW. Verify that simulated target pulse varies from 20 ±1.0 usec to 30 ±1.0 usec after signal on channel A.

4.6.3.2 Sum and difference doppler. - Connect test equipment items 1 through 9 as shown in Figure 2. Adjust signal generator item 4 to deliver Signal #l and signal generator item 5 to deliver Signal #2 as specified in Table I. Adjust item 5 to be slaved to item 4 and produce Signal #2 15 usec ±3 usec after Signal #1. Perform the following.

(a) 'Set S1 to position 2.

(b) Adjust R1 from fully CCW to fully CN. Verify that signals at channel A and B of oscilloscope are 2.5 KHz ±25 -Hz, not less than 3 volts p-p in phase when R1 is CCW and out of phase when R1 is CW.

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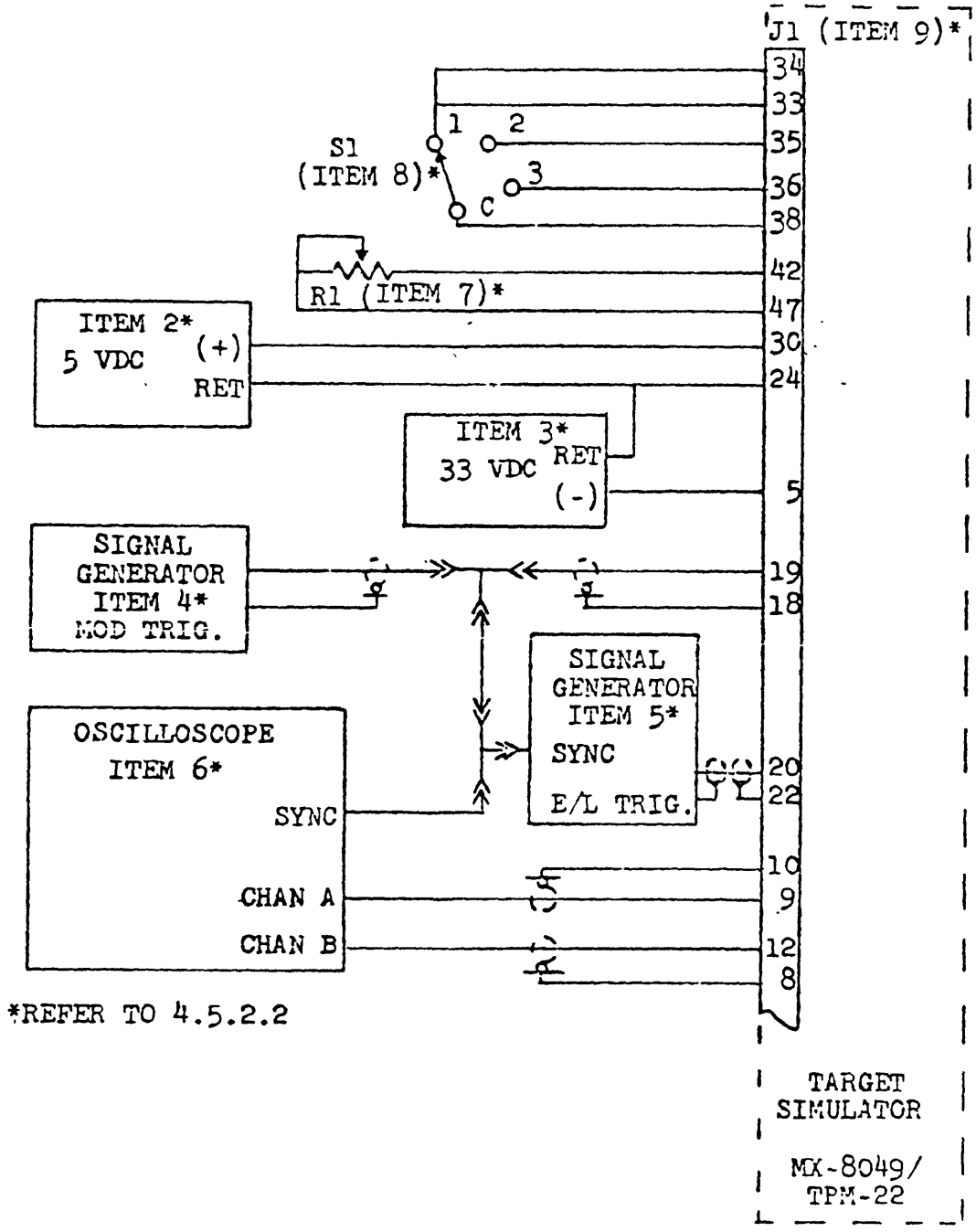


FIGURE 2. DOPPLER TEST SET UP

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10558244 The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The target simulator is used to provide a simulated target and doppler signals to enable the Range Computer Tester, Subassembly MX-8049/TPM-22, to locate defective portions of Unit 4 of Radar Set AN/VPS-2.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging arid packing.

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(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each target simulator.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 4931-A034

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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